SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	13/10/2026	**Duration:**	60 minutes

Code of Conduct
I S. LOHITHA - 192524031 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: *S. LOHITHA - 192524031*

Annexure A – Agile Sprint Planning & User Story Workshop

INTELLIGENT PARKING SPACE SHARING PLATFORM

1. Identify the Project Vision and Stakeholders

Project Vision

The vision of the Intelligent Parking Space Sharing Platform is to create a convenient digital platform that connects owners of unused parking spaces with drivers looking for temporary parking. The system enables drivers to find, reserve, and pay for suitable parking spaces while helping owners earn revenue from unused spaces.

Objectives

- Allow owners to list available parking spaces.
- Enable drivers to search and reserve parking spaces.
- Support secure digital payments and cancellations.
- Reduce the time spent searching for parking.
- Improve the utilization of unused parking spaces.
- Provide ratings, transaction history, and dispute management.

Key Stakeholders

1. **Parking-Space Owners** – List spaces, set prices, and manage availability.
2. **Drivers** – Search, reserve, pay for, and review parking spaces.
3. **System Administrator** – Manages users, listings, complaints, and disputes.
4. **Payment Service Provider** – Processes payments and refunds.
5. **Property/Security Personnel** – Manage access to restricted parking areas.
6. **Development Team** – Develops, tests, and maintains the platform.

2. Create Epics and User Stories

Epic 1: User Management

User Story 1:
As a driver, I want to register and log in, so that I can search and reserve parking spaces.

User Story 2:
As an owner, I want to create an account, so that I can list my available parking spaces.

Epic 2: Parking Space Management

User Story 3:
As an owner, I want to add parking-space details, so that drivers can find and reserve my space.

User Story 4:
As an owner, I want to update availability and pricing, so that drivers receive accurate information.

Epic 3: Search and Reservation

User Story 5:
As a driver, I want to search parking spaces by location, price, vehicle type, and availability, so that I can find a suitable space.

User Story 6:
As a driver, I want to reserve an available parking space, so that parking is confirmed before I arrive.

Epic 4: Payment and Cancellation

User Story 7:
As a driver, I want to make a digital payment, so that I can securely confirm my reservation.

User Story 8:
As a driver, I want to cancel my reservation, so that I can receive an eligible refund.

Epic 5: Ratings and Disputes

User Story 9:
As a driver, I want to rate and review a parking space, so that other drivers can make better decisions.

User Story 10:
As a user, I want to report a problem, so that the administrator can resolve disputes.

3. Define Acceptance Criteria

User Story 1 – User Registration

Given a new user opens the registration page,
When valid details are entered and submitted,
Then the system should create the account successfully.

User Story 3 – Parking-Space Listing

Given an owner is logged in,
When the owner enters valid location, price, availability, and vehicle details,
Then the parking space should be added successfully.

User Story 5 – Search Parking

Given a driver enters search requirements,
When the search is performed,
Then the system should display matching available parking spaces.

User Story 6 – Reservation

Given a parking space is available,
When the driver selects the required date and time and confirms,
Then the system should create the reservation without overlapping another confirmed booking.

User Story 7 – Payment

Given a driver has selected a parking space,
When the payment is successfully completed,
Then the system should confirm the payment and reservation.

User Story 8 – Cancellation

Given a driver has an active reservation,
When the driver cancels it,
Then the system should update the booking status and process any eligible refund.

4. Prioritize the Product Backlog

The product backlog is prioritized based on business value, user needs, and feature dependencies.

Priority	Product Backlog Item	Priority Level
1	User Registration and Login	High
2	Parking-Space Listing	High
3	Search and Filtering	High
4	Reservation Management	High
5	Digital Payment	High
6	Cancellation and Refund	Medium
7	Notifications	Medium
8	Ratings and Reviews	Medium
9	Dispute Management	Low
10	AI Recommendations	Low

Registration and parking listings receive the highest priority because other features depend on them. Search, reservations, and payments are prioritized next because they form the main parking-booking process.

5. Plan the Sprint Backlog

Sprint 1 Duration: 2 Weeks

The first sprint will focus on creating the basic system required for owners and drivers to start using the platform.

User Story	Development Tasks
User Registration	Design registration page, create database, validate user details
User Login	Create login page, authentication and logout
Owner Profile	Create owner profile and profile editing
Driver Profile	Create driver profile and vehicle details
Parking Listing	Create add, edit, and view parking-space functions

At the end of Sprint 1, owners and drivers should be able to create accounts and owners should be able to add basic parking-space listings.

6. Estimate Effort Using Story Points

The **Fibonacci estimation technique** is used with story points of 1, 2, 3, 5, 8, etc.

User Story	Story Points	Justification
User Registration	3	Basic form and validation
User Login	3	Requires authentication
Owner Profile	2	Simple profile management
Driver & Vehicle Profile	3	Requires vehicle information
Parking-Space Listing	5	Requires multiple details and database operations

Total Sprint 1 Story Points: 16

Parking-space listing receives more story points because it requires location, price, availability, vehicle restrictions, and database operations. Simpler profile-management functions require fewer story points.

7. Present the Sprint Goal and Expected Deliverables

Sprint Goal

The goal of Sprint 1 is to develop the basic user and parking-listing functions required to establish the foundation of the Intelligent Parking Space Sharing Platform.

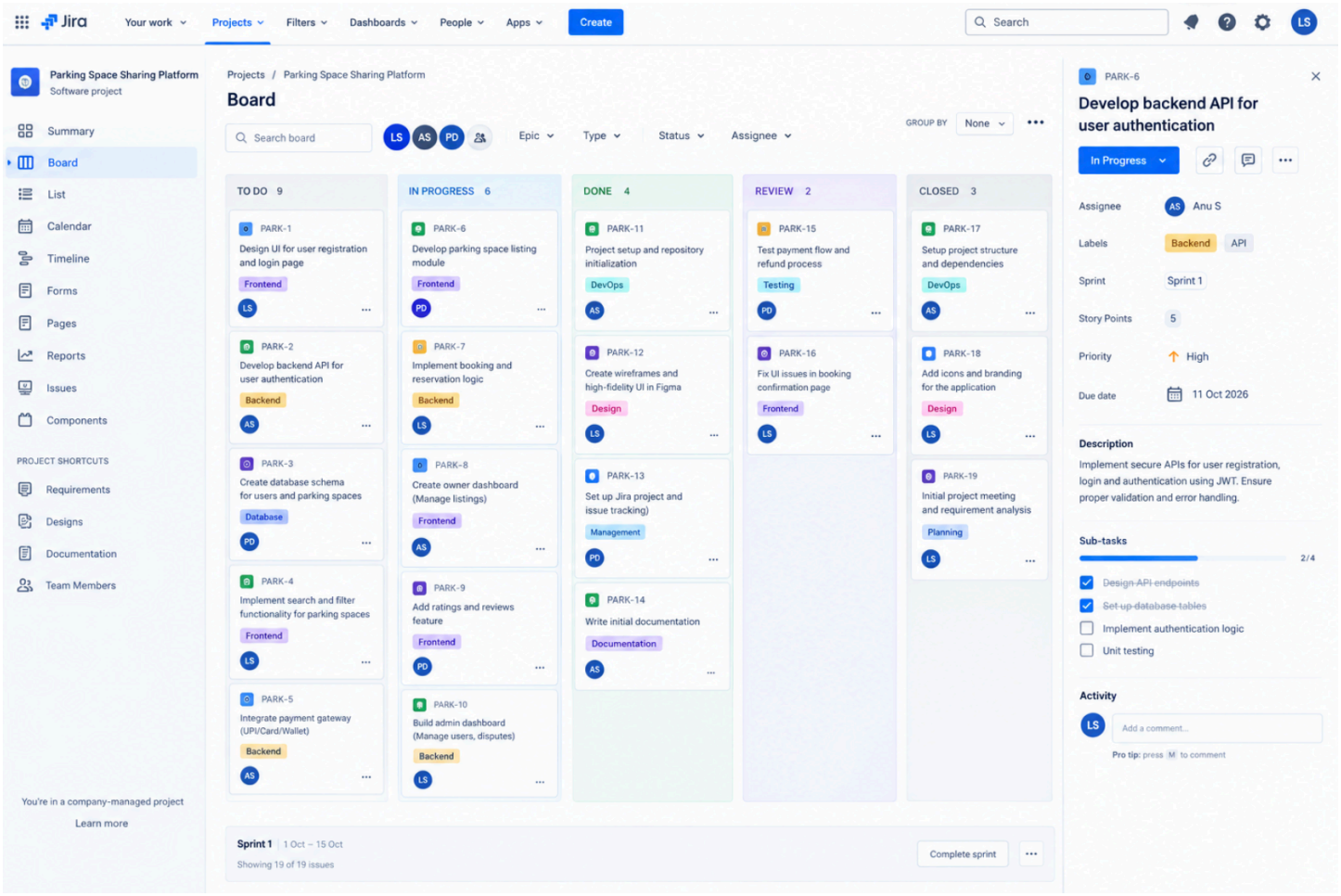
Expected Deliverables

At the end of Sprint 1, the following features should be completed:

- Owner and driver registration.
- Secure user login and logout.
- Owner profile management.
- Driver and vehicle profile management.
- Add, edit, and view parking-space listings.
- Basic database for users, vehicles, and parking spaces.
- Tested and working Sprint 1 prototype.

Conclusion

The Agile Sprint Planning approach divides the Intelligent Parking Space Sharing Platform into manageable epics, user stories, and development tasks. Prioritizing essential features such as registration, parking listings, search, reservations, and payments allows the platform to be developed step by step. Story-point estimation and sprint planning help the team manage workload and deliver a functional system efficiently.


Jira
Your work
Projects
Filters
Dashboards
People
Apps
Create
Search
LS
Parking Space Sharing Platform
Software project
Summary
Board
List
Calendar
Timeline
Forms
Pages
Reports
Issues
Components
PROJECT SHORTCUTS
Requirements
Designs
Documentation
Team Members
You're in a company-managed project
Learn more
Projects / Parking Space Sharing Platform
Board
Search board
LS AS PD
Epic
Type
Status
Assignee
GROUP BY None
TO DO 9
PARK-1
Design UI for user registration and login page
Frontend
PARK-2
Develop backend API for user authentication
Backend
PARK-3
Create database schema for users and parking spaces
Database
PARK-4
Implement search and filter functionality for parking spaces
Frontend
PARK-5
Integrate payment gateway (UPI/Card/Wallet)
Backend
IN PROGRESS 6
PARK-6
Develop parking space listing module
Frontend
PARK-7
Implement booking and reservation logic
Backend
PARK-8
Create owner dashboard (Manage listings)
Frontend
PARK-9
Add ratings and reviews feature
Frontend
PARK-10
Build admin dashboard (Manage users, disputes)
Backend
DONE 4
PARK-11
Project setup and repository initialization
DevOps
PARK-12
Create wireframes and high-fidelity UI in Figma
Design
PARK-13
Set up Jira project and issue tracking)
Management
PARK-14
Write initial documentation
Documentation
REVIEW 2
PARK-15
Test payment flow and refund process
Testing
PARK-16
Fix UI issues in booking confirmation page
Frontend
CLOSED 3
PARK-17
Setup project structure and dependencies
DevOps
PARK-18
Add icons and branding for the application
Design
PARK-19
Initial project meeting and requirement analysis
Planning
Sprint 1 | 1 Oct – 15 Oct
Showing 19 of 19 issues
Complete sprint
PARK-6
Develop backend API for user authentication
In Progress
Assignee Anu S
Labels Backend API
Sprint Sprint 1
Story Points 5
Priority High
Due date 11 Oct 2026
Description
Implement secure APIs for user registration, login and authentication using JWT. Ensure proper validation and error handling.
Sub-tasks
2/4
Design API endpoints
Set up database tables
Implement authentication logic
Unit testing
Activity
Add a comment...
Pro tip: press M to comment

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25